

👉 XAI Lecture 12

Robust & Probabilistically Robust Algorithmic Recourse

FAMAF - UNC

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

TOWARDS ROBUST AND RELIABLE ALGORITHMIC RECOURSE

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ABSTRACT

As predictive models are increasingly being deployed in high-stakes decision making (e.g., loan approvals), there has been growing interest in post-hoc techniques which provide recourse to affected individuals. These techniques generate recourses under the assumption that the underlying predictive model does not change. However, in practice, models are often regularly updated for a variety of reasons (e.g., dataset shifts), thereby rendering previously prescribed recourses ineffective. To address this problem, we propose a novel framework, RObust Algorithmic Recourse (ROAR), that leverages adversarial training for finding recourses that are robust to model shifts. To the best of our knowledge, this work proposes the first ever solution to this critical problem. We also carry out detailed theoretical analysis which underscores the importance of constructing recourses that are robust to model shifts: 1) we derive a lower bound on the probability of invalidation of recourses generated by existing

(Upadhyay, Joshi, and Lakkaraju 2021)

Motivation ---

Algorithmic Recourse

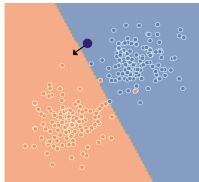
- ML models are deployed in high stakes scenarios
- If you receive an unfavorable outcome as a result of a prediction, how can you reverse it?
- Ex: A bank might tell you to increase your salary by \$10,000

Model Updates

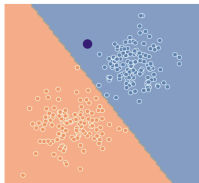
- In practice, data collectors are (hopefully) frequently updating their datasets
- Models are updated to reflect dataset changes
- Current algorithms to generate counterfactuals **assume models are static**

🍷 Motivation: An Example ---

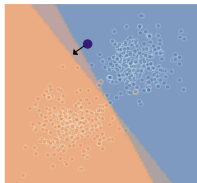
Model
trained on
 data_1



Model
trained on
 data_2



Overlaid



Original suggested recourse for the datapoint no longer crosses the decision boundary when the model is updated/retrained

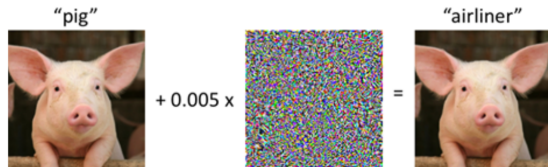
Summary of Contributions ---

- **Outline model + data shifts** that people should consider
 - ▶ Temporal shift
 - ▶ Geospatial shift
 - ▶ Data correction shift
- **Propose ROAR** — RObust Algorithmic Recourse
 - ▶ Introduces a novel minimax objective that can be used to construct robust actionable recourses while minimizing the recourse costs
- **Theoretical analysis**
 - ▶ How bad are regular counterfactuals under model shifts?
 - ▶ How much does the proposed method increase the cost of recourses when compared to normal CFs?
- **Experimental analysis**

🍷 Related Work ---

Algorithmic recourse

Can i still trust you?: Understanding the impact of distribution shifts on algorithmic recourses: recourses generated by state-of-the-art algorithms are readily invalidated due to model shifts resulting from different kinds of dataset shifts



Adversarial Training

- Optimizes a **minimax objective** that captures the worst-case loss over a given set of perturbations to the input data
- At each gradient step, **computes the gradient at worst-case perturbation**
- Recent work explores robust feature attribution and rule-based explanations robust to *dataset* shifts

In plain English

Small, carefully chosen perturbations can break a system — be it a classifier or a recourse recommendation.

Background: Recourse/Counterfactuals ---

Optimal CF:

$$x' = \arg \min_{x' \in \mathcal{A}} c(x, x') \quad \text{s.t.} \quad \mathcal{M}(x') = 1$$

Unconstrained and differentiable relaxation:

$$x' = \arg \min_{x' \in \mathcal{A}} \ell(\mathcal{M}(x'), 1) + \lambda c(x, x')$$

Notation Notes

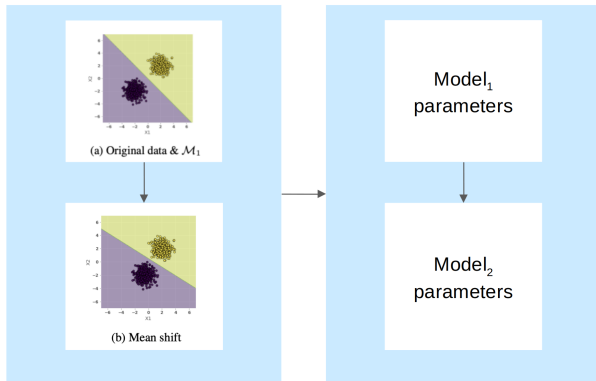
- x : specific data point
- x' : counterfactual
- $\mathcal{M}$: model (or linear approximation of model around point x)
- C : cost function (how hard is it to achieve the counterfactual)
- $\mathcal{A}$: actionable/possible counterfactuals

In plain english

Find the cheapest change to your profile that flips the model's decision. The second formulation makes this optimizable.



A Primer: Comparing Models Under Shifts ---



A mean shift occurs when the average feature values of one or more classes move, causing the decision boundary to rotate and previously valid recourses to fail.

Key idea

How to compare these models? (especially if they are nonlinear and high-dimensional)

Approach (Intuition) ---

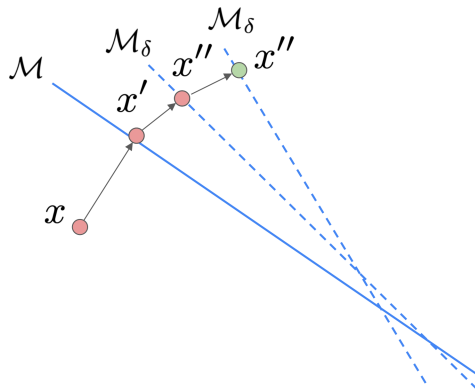
Adversarial Training

Perturb inputs slightly (maximize training loss) → gradient descent step on perturbed inputs (minimize training loss)

ROAR

Perturb model slightly (maximize recourse loss) → gradient descent step on perturbed model (minimize recourse loss)

🍷 Approach (Intuition): Diagram ---



The Goal

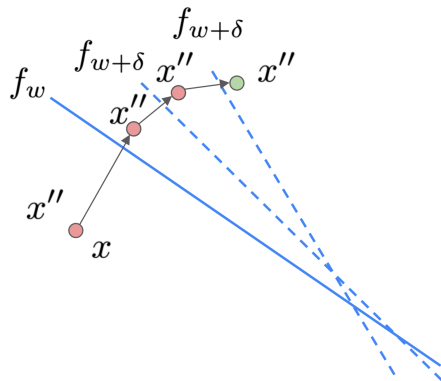
Find x'' that remains valid under all model perturbations $\delta \in \Delta$

🍷 Approach (Details): Algorithm Walkthrough ---

The algorithm starts from $x'' = x$ (your current profile) and repeats two steps until convergence:

- **Step 1 — Attack:** fix x'' and find the $\hat{\delta}$ that damages it most — the shift most likely to push it back to the negative side of the boundary.
- **Step 2 — Defend:** fix $\hat{\delta}$ and move x'' via gradient descent to survive that worst case, without straying too far from x (the term $\lambda c(x'', x)$ controls the cost).

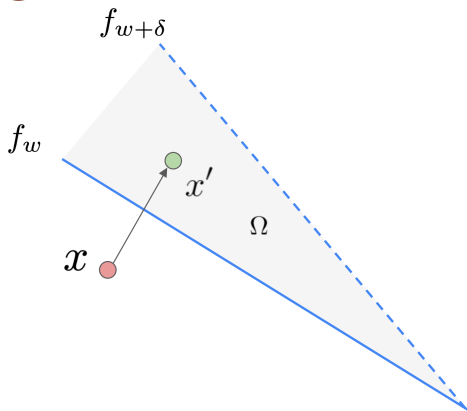
The diagram shows exactly this: x'' starts at x , crosses f_w in the first step, then keeps moving until it lands in the green zone — valid under all possible $f_{w+\delta}$.



In plain English

Alternate between “find the worst model shift” and “move the recourse to survive it.”

🍷 Proofs: Theorem 1 ---



$$\Omega = \{x' : w^T x' > 0 \cap (w + \delta)^T x' \leq 0\}$$

... integrating over Ω

$$P(x' \text{ is invalidated}) \geq \frac{1}{2} \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta - 1}}{\beta} \exp \left(-\beta \frac{(w^T \mu)^2}{2 \|\sqrt{D} U w\|^2} \right)$$

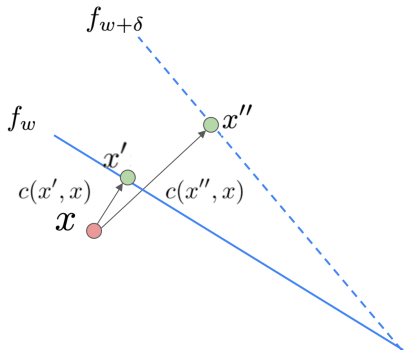
Assumptions: $x \sim \mathcal{N}(\mu, \Sigma)$, $x' \sim \mathcal{N}(\mu, \Sigma)$,
 $\Sigma = U D U^T$, $\beta \geq 1$

The theorem answers: how likely is a standard (non-robust) recourse to fail under a model shift? (Ω is the “danger zone”)

In plain English

If you use a standard recourse, there's a provably non-negligible chance it gets invalidated when the model updates — and this theorem quantifies that floor.

🍷 Proofs: Theorem 2 ---



w.h.p. $(1 - \eta')$:

$$c(x'', x) \leq c(x', x) + \frac{1}{\lambda \|w + \delta\|} \alpha(w + \delta)^T \mu + \sqrt{\frac{D^2}{2} \log \frac{1}{\eta'}}$$

Assumptions:

- Log-loss + simple model
- Optimal solution for x''
- D is a bound for the “diameter” of dataset (I_2)

The theorem answers: **how much more expensive is the robust recourse compared to the standard one?**

In plain English

Robust recourse costs more than standard recourse, but the extra cost is bounded — and the bound grows with how much you allow the model to shift.

Experimental Results ---

Real World Datasets

- German Credit → Data Correction Shift
- Small Business Administration → Temporal Shifts
- Portuguese Student Performance → Geospatial Shift

Synthetic Datasets

- 2 Gaussians: Mean Shift, Variance Shift, Mean & Variance Shift

Models: LR, SVM, 3 Layer Deep NN

Cost Functions:

- L1
- Pairwise Feature Comparison (PFC):
Bradley Terry Comparison to parametrize $p(i, j)$ = probability that feature i is less actionable than feature j

Metrics:

- **Cost:** How close is our counterfactual?
- **Validity:** When undertaking recourse (after a model shift) does it actually work?

Baselines ---

Acronym	Full name	Key idea
CFE	Counterfactual Explanations (Wachter et al.)	Closest point that flips the prediction
AR	Actionable Recourse (Ustun et al.)	Integer programming, only actionable features
CR	Causal Recourse / MINT (Karimi et al.)	Respects causal structure of features
ROAR	Robust Algorithmic Recourse (Upadhyay et al.)	Minimax objective, robust to model shifts

In plain English

CFE and AR find cheap recourses but ignore model updates. CR adds causal structure. ROAR is the only method designed to survive model shifts.

Types of Dataset Shift ---

Shift	Description	Dataset
Correction shift	Two versions of the same dataset where the second corrects coding errors in the first	German Credit
Temporal shift	Data distribution changes over time	Small Business Administration (1989–2012)
Geospatial shift	Data distribution changes across geographic locations	Portuguese Student Performance (2 schools)

In plain English

Each shift type simulates a realistic reason why a deployed model gets retrained — and therefore why a previously valid recourse might fail.

🍷 Experimental Results: Logistic Regression 1/3 ---

			Correction Shift		
Model	Cost	Recourse	Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
LR	L1	CFE	0.74 ± 0.08	1.00 ± 0.00	0.36 ± 0.08
		AR	0.72 ± 0.06	1.00 ± 0.00	0.48 ± 0.12
		ROAR	2.80 ± 0.07	0.96 ± 0.04	0.89 ± 0.08
		CR	3.45 ± 0.64	1.00 ± 0.00	0.86 ± 0.09
	PFC	CFE	0.02 ± 0.01	1.00 ± 0.00	0.45 ± 0.13
		AR	0.07 ± 0.01	1.00 ± 0.00	0.49 ± 0.10
		ROAR	0.31 ± 0.02	1.00 ± 0.00	1.00 ± 0.00
		CR	1.04 ± 0.55	1.00 ± 0.00	0.89 ± 0.08

$$\arg \min_{x'' \in \mathcal{A}} \max_{\delta \in \Delta} \ell(\mathcal{M}_\delta(x''), 1) + \lambda c(x, x'') \quad \arg \min_{x'} \max_{\lambda} \lambda (f_w(x') - y')^2 + d(x_i, x')$$

🍷 Experimental Results: Logistic Regression 2/3 ---

Model	Cost	Recourse	Temporal Shift		
			Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
LR	L1	CFE	3.46 ± 0.31	1.00 ± 0.00	0.35 ± 0.06
		AR	1.53 ± 0.09	1.00 ± 0.00	0.27 ± 0.05
		ROAR	3.65 ± 0.21	1.00 ± 0.00	0.96 ± 0.01
		CR	NA	NA	NA
	PFC	CFE	0.22 ± 0.02	1.00 ± 0.00	0.27 ± 0.08
		AR	0.12 ± 0.01	1.00 ± 0.00	0.14 ± 0.04
		ROAR	0.56 ± 0.08	1.00 ± 0.00	0.96 ± 0.01
		CR	NA	NA	NA

CR/MINT requires a known causal graph as input — since this was only available for the German Credit dataset, results for temporal and geospatial shifts are NA.

🍷 Experimental Results: Logistic Regression 3/3 ---

			Temporal Shift		
Model	Cost	Recourse	Geospatial Shift		
LR	L1	CFE	Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
		AR	8.44 ± 0.30	1.00 ± 0.00	0.23 ± 0.04
		ROAR	5.24 ± 0.21	1.00 ± 0.00	0.35 ± 0.11
		CR	11.07 ± 0.48	1.00 ± 0.00	0.68 ± 0.05
	PFC	CFE	NA	NA	NA
		AR	0.34 ± 0.03	1.00 ± 0.00	0.18 ± 0.04
		ROAR	0.32 ± 0.02	1.00 ± 0.00	0.23 ± 0.04
		CR	1.13 ± 0.03	1.00 ± 0.00	0.88 ± 0.06
			NA	NA	NA

CR/MINT requires a known causal graph as input — since this was only available for the German Credit dataset, results for temporal and geospatial shifts are NA.

🍷 Experimental Results: Deep NN 1/3 ---

Correction Shift

			Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
NN	L1	CFE	0.52 ± 0.05	1.00 ± 0.00	0.48 ± 0.05
		AR-LIME	0.37 ± 0.06	0.19 ± 0.06	0.36 ± 0.07
		ROAR-LIME	1.90 ± 0.15	0.75 ± 0.01	0.84 ± 0.04
		CR	2.80 ± 0.59	0.85 ± 0.02	0.74 ± 0.02
	PFC	CFE	0.05 ± 0.01	1.00 ± 0.00	0.44 ± 0.06
		AR-LIME	0.05 ± 0.01	0.34 ± 0.08	0.38 ± 0.04
		ROAR-LIME	0.59 ± 0.05	0.87 ± 0.02	0.84 ± 0.03
		CR	0.64 ± 0.08	0.88 ± 0.03	0.72 ± 0.02

🍷 Experimental Results: Deep NN 2/3 ---

Temporal Shift

			Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
NN	L1	CFE	5.93 ± 1.17	1.00 ± 0.00	0.61 ± 0.07
		AR-LIME	1.70 ± 0.11	0.54 ± 0.08	0.77 ± 0.04
		ROAR-LIME	5.42 ± 0.26	0.99 ± 0.01	0.97 ± 0.01
		CR	NA	NA	NA
	PFC	CFE	0.37 ± 0.10	1.00 ± 0.00	0.48 ± 0.06
		AR-LIME	0.14 ± 0.01	0.48 ± 0.06	0.73 ± 0.05
		ROAR-LIME	0.59 ± 0.14	0.996 ± 0.003	0.99 ± 0.01
		CR	NA	NA	NA

🍷 Experimental Results: Deep NN 3/3 ---

Geospatial Shift

			Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
NN	L1	CFE	0.50 ± 0.05	1.00 ± 0.00	0.46 ± 0.05
		AR-LIME	9.50 ± 0.50	1.00 ± 0.00	0.46 ± 0.05
		ROAR-LIME	6.99 ± 0.29	0.60 ± 0.05	0.76 ± 0.08
		CR	18.65 ± 2.07	0.997 ± 0.003	0.98 ± 0.01
	PFC	CFE	NA	NA	NA
		AR-LIME	0.44 ± 0.04	1.00 ± 0.00	0.36 ± 0.09
		ROAR-LIME	0.60 ± 0.09	0.57 ± 0.06	0.53 ± 0.05
		CR	1.59 ± 0.15	1.00 ± 0.00	0.96 ± 0.01
			NA	NA	NA

Takeaways ---

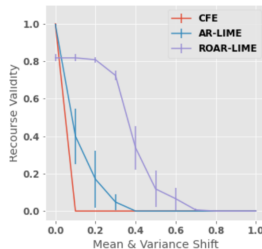
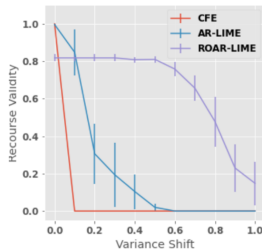
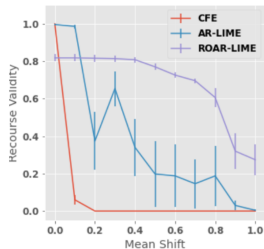
- Cost increases with robust recourse in general
 - ▶ The authors bound this!
- Linear approximations to complex models harms $\mathcal{M}_1$ validity
 - ▶ Robustness can actually help!
- We can optimize for $\mathcal{M}_1$ validity by making our loss function more complex:

$$\arg \min_{x''} \max_{\delta} \max_{\lambda} \lambda \ell(M(x''), 1) + c(x, x'')$$

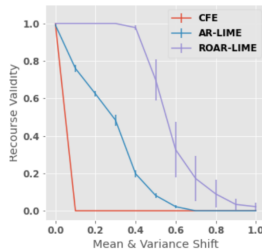
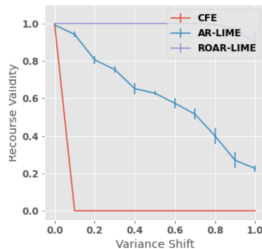
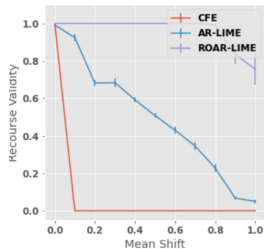


Experimental Results: Shift and Validity ---

L1:



PFC:



🍷 Conclusions (Paper 1) ---

- Novel minimax objective and optimization strategy
- Bounds on error for non-robust counterfactuals under data shifts
- Bounds on cost of robust optimization
- Empirical results in both real world and synthetic scenarios validating method

Questions/Discussion ---

- 1 Do you think this problem should be worked on more from the CS/ML side or more from the policy side?
- 2 Do you think other types of explanations (that are not recourse motivated) need to be similarly robust?
- 3 What happens if model architecture/training dynamics change?
- 4 What incentive is there for companies/data controllers to implement this?

Published as a conference paper at ICLR 2023

PROBABILISTICALLY ROBUST RECOURSE: NAVIGATING THE TRADE-OFFS BETWEEN COSTS AND ROBUSTNESS IN ALGORITHMIC RECOURSE

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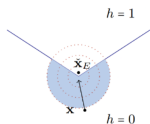
ABSTRACT

As machine learning models are increasingly being employed to make consequential decisions in real-world settings, it becomes critical to ensure that individuals who are adversely impacted (e.g., loan denied) by the predictions of these models are provided with a means for recourse. While several approaches have been proposed to construct recourse for affected individuals, the recourse output by

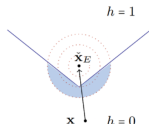
(Pawelczyk et al. 2022)

🍷 Motivation (Paper 2) ---

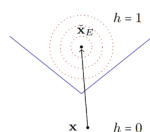
- Algorithmic recourse aims to provide users with actionable changes to move from a negative to a positive prediction in a ML model
- Typically, the minimum cost change is computed
- In the real-world, these changes are often **implemented noisily**
 - ▶ E.g. an individual who was asked to increase their salary by \$500 may get a promotion which comes with a raise of \$505 or even \$499.95
 - ▶ Or, changing certain factors may cause others to change



(a) Low recourse cost and low recourse robustness



(b) Medium recourse cost and medium recourse robustness



(c) High recourse cost and high recourse robustness

🍷 Related Work (Paper 2) ---

- **Counterfactual Explanations and Recourse**

- ▶ A plethora of papers including Wachter et al. (2018) (Wachter, Mittelstadt, and Russell 2018)
- ▶ Summarised in Verma et al. (2020) (Verma, Dickerson, and Hines 2020):
 - ★ Type of the underlying predictive model
 - ★ Whether they encourage sparsity in counterfactuals
 - ★ Whether counterfactuals should lie on the data manifold
 - ★ Whether causal relationships should be accounted for

- **Robustness to model/data shift:**

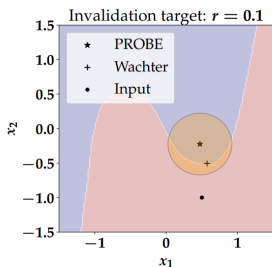
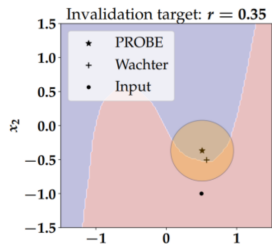
- ▶ ROAR — Robust Algorithmic Recourse, Upadhyay et al. (2021) (Upadhyay, Joshi, and Lakkaraju 2021)

- **Robustness to input perturbations:**

- ▶ Causal Algorithmic Recourse, Dominguez-Olmedo et al. (2022) (Dominguez-Olmedo, Karimi, and Schölkopf 2022)

- All approaches generate recourses **assuming that prescribed recourses will be correctly implemented** by users

Solution: PROBE ---



- **PROBE** — Probabilistically Robust Recourse
- Allows users to manage the **recourse cost vs. robustness tradeoffs**
- Users can choose the **recourse invalidation rate**
 - ▶ Probability with which a recourse could get invalidated
- PROBE recourses are, compared to baselines:
 - ▶ Less costly than previous methods
 - ▶ More robust to noisy implementations
 - ▶ Able to provide recourses at various invalidation rates

Notation ---

$$\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$$

input space

$$\mathcal{Y} = \{0, 1\}$$

output space

- 0: unfavourable outcome
- 1: favourable outcome

$$h : \mathcal{X} \rightarrow \mathcal{Y}$$

classifier

$$h(\mathbf{x}) = g(f(\mathbf{x}))$$

$$f : \mathcal{X} \rightarrow \mathbb{R}$$

inputs $\rightarrow$ logits

$$g : \mathbb{R} \rightarrow \mathcal{Y}$$

logits $\rightarrow$ binary labels

In plain English

The classifier is split into f (continuous score) and g (decision threshold) because PROBE needs to reason about *how far* a noisy implementation lands from the boundary — not just which side it lands on.

🍷 General Formulation of Algorithmic Recourse ---

$$\check{\mathbf{x}} = \arg \min_{\mathbf{x}' \in \mathcal{A}} \underbrace{\ell(h(\mathbf{x}'), 1)}_{\text{valid outcome}} + \lambda \cdot \underbrace{d_c(\mathbf{x}, \mathbf{x}')}_{\text{low cost}}$$

This is the standard formulation — it optimizes for validity and cost, but assumes the individual implements the recourse **exactly as recommended**.

What's missing: no term accounts for noise in the implemented counterfactual. PROBE addresses this by replacing the validity term with one that explicitly models implementation uncertainty.

Recourse Invalidation Rate ---

$$\Delta(\check{\mathbf{x}}_E) = \mathbb{E}_{\varepsilon} \left[\underbrace{h(\check{\mathbf{x}}_E)}_{\text{CF class}} - \underbrace{h(\check{\mathbf{x}}_E + \varepsilon)}_{\text{class after response}} \right]$$

where $\varepsilon \sim p_{\varepsilon} \rightarrow$ probability distribution that captures noise in response

e.g. $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

In plain English

$\Delta(\check{\mathbf{x}}_E)$ measures the average fraction of times the recourse fails due to implementation noise — zero means always valid, one means always fails.

Recourse Invalidation Rate Aware Objective ---

$$\mathcal{L} = \underbrace{R(\mathbf{x}'; r, \sigma^2 \mathbf{I})}_{\text{make IR of CF close to target IR (new term)}} + \underbrace{\ell(f(\mathbf{x}'), s)}_{\text{make CF have favourable outcome}} + \underbrace{\lambda d_c(\mathbf{x}', \mathbf{x})}_{\text{low cost}}$$

where

$$R(\mathbf{x}'; r, \sigma^2 \mathbf{I}) = \max(0, \underbrace{\Delta(\mathbf{x}'; \sigma^2 \mathbf{I})}_{\text{CF's IR}} - \underbrace{r}_{\text{target IR}})$$

- The three terms together read: find a resource that is valid, cheap, and whose invalidation rate does not exceed **R**
- **PROBE** adds a single knob **R**: the user-specified maximum failure rate, and penalizes any recourse that exceeds it, leaving the standard cost/validity tradeoff untouched otherwise

🍷 Approximation of Recourse Invalidation Rate (Theorem 1) ---

Problem: $\Delta(\mathbf{x}')$ is not differentiable (used in objective function)

Solution: use a first order approximation of $\Delta(\mathbf{x}')$

$$\tilde{\Delta}(\check{\mathbf{x}}_E; \sigma^2 \mathbf{I}) = 1 - \Phi \left(\frac{f(\check{\mathbf{x}}_E)}{\sqrt{\nabla f(\check{\mathbf{x}}_E)^\top \sigma^2 \mathbf{I} \nabla f(\check{\mathbf{x}}_E)}} \right)$$

where $\check{\mathbf{x}}_E$: counterfactual, $f(\check{\mathbf{x}}_E)$: logit at counterfactual

Proof sketch:

- 1 Solve $\mathbb{P}(f(\check{\mathbf{x}}_E + \varepsilon) > 0)$
- 2 Use first order Taylor series to approximate logit
- 3 Calculate probability that normal r.v. is less than a value $\rightarrow$ CDF

In plain English

Δ is not differentiable, so PROBE linearizes the model around $\check{\mathbf{x}}_E$ and reduces the problem to a Gaussian CDF — cheap to compute and differentiable.

Theorem 1: Further Analyses ---

- **Proposition 1** — Wachter et al. (2018) method for logistic regression
 - ▶ Derive closed form solution for IR of CF
 - ▶ Show how to make CF more robust
- **Proposition 2** — PROBE recourse incurs an additional cost (linear regression)
- **Proposition 3** — Upperbound on IR

In plain English

These three results mirror ROAR's theorems: robustness is achievable, it costs more but not unboundedly so, and the failure rate is provably controlled.

Experimental Evaluation: Datasets ---

- **Adult Dataset:** Predict whether an individual has an income greater than 50,000 USD/year
- **Give Me Some Credit:** Predict whether an individual will experience financial distress within the next two years or not
- **COMPAS:** Predict if criminal is high or low risk of re-offending



Experimental Evaluations: Baselines ---

- **Baseline Methods**

- ▶ **Growing Spheres (GS):** Random search algorithm — generate observations until decision boundary is crossed then move greedily toward decision boundary
- ▶ **AR (-LIME):** Actionable Recourse in Linear Models → use integer programming
- ▶ **DICE:** Diverse Counterfactual Explanations — promote diversity of counterfactual explanations
- ▶ **Gradient:** General formulation of algorithmic recourse

- **Adversarial Minmax Objectives Methods**

- ▶ **ROAR:** Recourse robust to model changes by generating CFs that minimize worst-case loss over plausible model shifts
- ▶ **ARAR:** Adversarial Robustness of Causal Algorithmic Recourse — recourse robust to features of individual

Experimental Evaluation: Measures Used ---

- **Average Cost (AC):** Average cost for all prescribed recourses for the test set (implemented as l1-norm)
- **Recourse Accuracy (RA):** For all prescribed recourses what fraction results in the desired prediction
- **Average IR (AIR):** Average recourse invalidation rate for all prescribed recourses in test set



Experimental Evaluation: Results ---

	Measures	Adult				Compas				GMC			
		AR	Wachter	GS	PROBE	AR	Wachter	GS	PROBE	AR	Wachter	GS	PROBE
LR	RA (↑)	0.98	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	AIR (↓)	0.5 ± 0.01	0.46 ± 0.02	0.35 ± 0.11	0.34 ± 0.02	0.48 ± 0.04	0.47 ± 0.02	0.3 ± 0.18	0.28 ± 0.02	0.47 ± 0.06	0.45 ± 0.03	0.48 ± 0.04	0.24 ± 0.01
	AC (↓)	0.55 ± 0.4	0.62 ± 0.43	2.12 ± 1.05	1.56 ± 0.92	0.16 ± 0.17	0.22 ± 0.17	0.73 ± 0.45	0.63 ± 0.39	0.29 ± 0.27	0.49 ± 0.51	0.28 ± 0.31	0.60 ± 0.56
NN	RA (↑)	0.38	1.0	1.0	0.99	0.84	1.0	1.0	1.0	0.38	1.0	1.0	1.0
	AIR (↓)	0.49 ± 0.03	0.5 ± 0.02	0.48 ± 0.02	0.35 ± 0.01	0.34 ± 0.09	0.46 ± 0.02	0.43 ± 0.07	0.33 ± 0.02	0.34 ± 0.07	0.43 ± 0.03	0.45 ± 0.03	0.25 ± 0.03
	AC (↓)	1.05 ± 0.22	0.3 ± 0.19	2.99 ± 1.51	1.43 ± 0.49	1.15 ± 0.52	0.2 ± 0.16	0.81 ± 0.45	0.8 ± 0.34	0.2 ± 0.19	0.26 ± 0.18	0.12 ± 0.09	0.47 ± 0.21

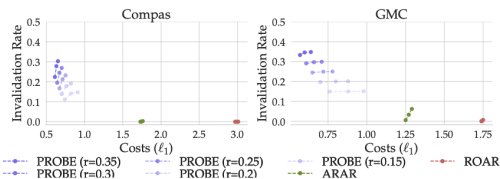
(a) Comparing PROBE to baseline recourse methods.

	Measures	Adult			Compas			GMC		
		ROAR	ARAR	PROBE	ROAR	ARAR	PROBE	ROAR	ARAR	PROBE
LR	RA(↑)	1.0	1.0	1.0	1.0	0.99	1.0	1.0	1.0	1.0
	AIR (↓)	0.0 ± 0.0	0.02 ± 0.01	0.34 ± 0.02	0.0 ± 0.0	0.0 ± 0.0	0.28 ± 0.02	0.0 ± 0.0	0.35 ± 0.01	0.24 ± 0.01
	AC (↓)	3.56 ± 0.8	2.68 ± 0.79	1.56 ± 0.92	2.99 ± 0.31	1.74 ± 0.3	0.63 ± 0.39	1.74 ± 0.45	1.27 ± 0.45	0.60 ± 0.56
NN	RA(↑)	0.94	0.03	0.99	0.97	0.02	1.0	0.06	0.06	1.0
	AIR (↓)	0.0 ± 0.0	0.51 ± 0.0	0.35 ± 0.01	0.01 ± 0.06	0.46 ± 0.0	0.33 ± 0.02	0.3 ± 0.21	0.45 ± 0.01	0.25 ± 0.03
	AC (↓)	19.8 ± 3.39	0.04 ± 0.0*	1.43 ± 0.49	6.41 ± 1.07	0.02 ± 0.0*	0.8 ± 0.34	0.67 ± 0.94	0.02 ± 0.0*	0.47 ± 0.21

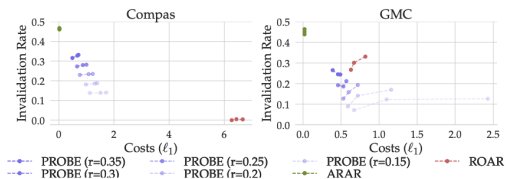
(b) Comparing PROBE to adversarially robust recourse methods.



Experimental Evaluation: Results (Cost vs. IR) ---



(a) Logistic Regression



(b) Neural Network

🍷 Conclusions & Future Work (Paper 2) ---

- First work to navigate tradeoff between recourse cost and robustness, give control to the user
- Experimental evidence demonstrates usefulness of framework and the existence of tradeoff
- Future work: generate recourse that is simultaneously robust to noise in inputs and shifts in model parameters

Discussions/Limitations ---

- “Generate recourse that is simultaneously robust to noise in inputs and shifts in model parameters”
 - ▶ Is there not a tradeoff still? Would the user have to provide different robustness thresholds for the different aspects the method is robust with respect to? Or is there one threshold to rule them all?
- Would users be okay with this risk of defining $r\%$ invalidation? Bias? How does recourse relate to bias?
- In general this idea of cost being described as l1 distance across methods seems to be a major limitation. Not all features incur similar contributions to cost (i.e. getting a degree versus putting more money into your savings account, one is harder than the other). Cost is a difficult concept to quantify.

Thank You!

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